



GHULAM ISHAQ KHAN INSTITUTE

of Engineering Sciences and Technology

Faculty of Computer Science and Engineering

CE342L • Computational Methods and Tools Lab

Lab Report #10

Power Method for Dominant Eigenvalue

Iterative Eigenvalue Computation in MATLAB

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Semester: 6th Semester



1

Objective	4
Theory	4
1 Task 01 — Power Method for 2×2 Matrix	6
1.1 Question	6
1.2 Theory	6
1.3 MATLAB Code	6
1.4 Output	7
2 Task 02 — Power Method for 3×3 Matrix	9
2.1 Question	9
2.2 Theory	9
2.3 MATLAB Code	9
2.4 Output	10
2.5 Comparison Table	10
Conclusion	12

1

The objective of this lab is to implement the **Power Method** in MATLAB to find the dominant eigenvalue and its corresponding eigenvector of a given square matrix. The Power Method is an iterative technique based on repeated matrix–vector multiplications that converges to the dominant eigenpair.

By the end of this lab, students will be able to:

- Understand the concepts of eigenvalues and eigenvectors.
- Define and identify the dominant eigenvalue of a matrix.
- Implement the Power Method with convergence checking in MATLAB.
- Analyse the effect of different initial guess vectors on convergence.

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► *Eigenvalues and Eigenvectors* For a square matrix $A_{n \times n}$, an eigenvalue λ and eigenvector $\mathbf{v}$ satisfy:

$$A\mathbf{v} = \lambda\mathbf{v}$$

The matrix A scales the special vector $\mathbf{v}$ by the factor λ without changing its direction. If $|\lambda_1| > |\lambda_i|$ for all $i \neq 1$, then λ_1 is called the **dominant eigenvalue** and $\mathbf{v}_1$ is the corresponding **dominant eigenvector**.

► *Algorithm Steps* **Step 1 — Initialization:** Choose a normalized initial guess $\mathbf{v}^{(0)}$ (largest component = 1).

Step 2 — Iteration:

$$\mathbf{w}^{(k)} = A\mathbf{v}^{(k-1)}, \quad \lambda^{(k)} = \max(|\mathbf{w}^{(k)}|), \quad \mathbf{v}^{(k)} = \frac{\mathbf{w}^{(k)}}{\lambda^{(k)}}$$

Step 3 — Convergence Check:

$$\|\mathbf{v}^{(k)} - \mathbf{v}^{(k-1)}\| < \varepsilon$$

Stop when the change in the eigenvector falls below tolerance ε .

► *Normalized Initial Guess* An initial guess is normalized if the largest magnitude component equals 1. For a 3×3 matrix, valid normalized initial guesses include:

$$[1; 1; 1], \quad [1; 1; 0], \quad [1; 0; 1], \quad [1; 0; 0]$$

► *Example (Manual)* — $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \mathbf{x}_0 = [1; 0]$

$$A\mathbf{x}_0 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}, \quad A\mathbf{x}_1 = \begin{bmatrix} 2.5 \\ 2 \end{bmatrix} = 2.5 \begin{bmatrix} 1 \\ 0.8 \end{bmatrix}, \quad \dots \quad A\mathbf{x}_6 = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Since $\mathbf{x}_6 \approx \mathbf{x}_7$, the dominant eigenvalue is $\lambda = 3$ and the eigenvector is $[1; 1]$.

```
► BaseMATLABImplementation(fromLab)
1 % Power Method to find dominant Eigenvector and Eigenvalue
2 A = input('Enter your matrix A:');
3 v = input('Enter normalized initial guess vector:');
4 n = input('Enter number of iterations:');
5
6 for i = 1:n
7     v = A * v;           % Update the vector v by multiplying with
8                           matrix A
9     M = max(v);          % Finding Maximum from vector v
10    v = v / M;           % Divide vector v with Maximum
11 end
```

Listing 1: Base Power Method provided in lab handout

1

1

► *Question* Find the dominant eigenvalue and eigenvector of the matrix

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

using the Power Method with:

- Initial guess: $v^{(0)} = [1; 0]$
- Maximum iterations: $n = 10$
- Tolerance: $\varepsilon = 10^{-4}$

Stop early if $\|v^{(k)} - v^{(k-1)}\| < \varepsilon$.

► *Theory* The analytical eigenvalues of A are found from $\det(A - \lambda I) = 0$:

$$(2 - \lambda)(3 - \lambda) - 1 = 0 \implies \lambda^2 - 5\lambda + 5 = 0 \implies \lambda = \frac{5 \pm \sqrt{5}}{2}$$

giving $\lambda_1 \approx 3.618$ (dominant) and $\lambda_2 \approx 1.382$. The dominant eigenvector corresponding to λ_1 is proportional to $[0.618; 1]$.

Parameter	Value
Matrix A	$\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$
Initial guess $v^{(0)}$	$[1; 0]$
Tolerance ε	10^{-4}
Max iterations	10

Table 1: Settings for Task 01

► *MATLABCode*

```

1 % AB #10 - Power Method (2x2 Matrix)
2 % Name: Daim Zak
3 % Reg No: 2023405
4
5 clc;
6 clear;
7
8 A = [2 1;
9      1 3];
10
11 v = [1; 0];           % initial guess
12 n = 10;               % max iterations
13 e = 1e-4;             % tolerance
14
15 lambda_old = 0;
16
17 for k = 1:n

```

```

1  % AB #10 - Power Method (3x3 Matrix)
2  % Name: Daim Zak
3  % Reg No: 2023405
4
5  clc;
6  clear;
7
8  A = [3 2 1;
9       2 3 2;
10      1 2 3];
11
12  n = 12;
13  e = 1e-5;
14
15  % Function for Power Method
16  power_method = @(v0) run_power(A, v0, n, e);
17
18  disp('--- Case A ---');
19  [vA, lambdaA, iterA] = power_method([1;0;0]);
20
21  disp('Eigenvalue:'); disp(lambdaA);
22  disp('Eigenvector:'); disp(vA);
23  disp('Iterations:'); disp(iterA);
24
25  disp('-----');
26
27  disp('--- Case B ---');
28  [vB, lambdaB, iterB] = power_method([1;1;1]);
29

```

Command Window

```

--- Case A ---
Eigenvalue:
    6.3723

Eigenvector:
    0.8431
    1.0000
    0.8431

Iterations:
    12

-----

--- Case B ---
Eigenvalue:
    6.3723

Eigenvector:
    0.8431
    1.0000
    0.8431

```

```

18  w      = A * v;           % matrix multiplication
19  lambda = max(abs(w));     % dominant eigenvalue estimate
20  v_new  = w / lambda;      % normalize vector
21
22  % convergence check
23  if norm(v_new - v) < e
24      fprintf('Converged at iteration %d\n', k);
25      break;
26  end
27
28  v      = v_new;
29  lambda_old = lambda;
30 end
31
32 disp('Dominant Eigenvalue:');
33 disp(lambda);
34 disp('Eigenvector:');
35 disp(v);

```

Listing 2: Task 01 — Power Method for 2×2 matrix

► *Output***Discussion**

The Power Method converges to the dominant eigenvalue $\lambda \approx \mathbf{3.6184}$ with corresponding eigenvector $\mathbf{v} \approx [0.6182; 1.0000]$. This matches the analytical value $\lambda_1 = (5 + \sqrt{5})/2 \approx 3.618$ closely. Early termination is triggered by the tolerance

condition, demonstrating efficient convergence; the algorithm does not need to run all 10 iterations. The eigenvector ratio $0.6182 \approx (\sqrt{5} - 1)/2$ confirms correctness.

2

1

► *Question* Apply the Power Method to find the dominant eigenvalue and eigenvector of

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

Run the algorithm for two different initial guesses and compare convergence:

$$\text{Case A: } v^{(0)} = [1; 0; 0], \quad \text{Case B: } v^{(0)} = [1; 1; 1]$$

with $n = 12$ iterations and tolerance $\varepsilon = 10^{-5}$.

► *Theory* For a symmetric matrix, all eigenvalues are real. The trace of A equals the sum of eigenvalues:

$$\text{tr}(A) = 3 + 3 + 3 = 9 = \lambda_1 + \lambda_2 + \lambda_3$$

The dominant eigenvalue is expected to be the largest of the three. Different normalized initial guesses affect how quickly the Power Method projects onto the dominant eigenvector subspace. If the initial guess happens to be orthogonal to the dominant eigenvector, convergence can stall; otherwise, both Case A and Case B should converge to the same λ_1 since the matrix is symmetric and the dominant eigenvector is unique (up to sign).

Parameter	Value
Matrix A	$\begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}$
Tolerance ε	10^{-5}
Max iterations	12
Case A initial guess	$[1; 0; 0]$
Case B initial guess	$[1; 1; 1]$

Table 2: Settings for Task 02

► *MATLAB Code*

```

1 % AB #10 - Power Method (3x3 Matrix)
2 % Name: Daim Zak
3 % Reg No: 2023405
4
5 clc;
6 clear;
7
8 A = [3 2 1;
9      2 3 2;
10     1 2 3];
11
12 n = 12;
13 e = 1e-5;
14
15 % Function for Power Method
16 power_method = @(v0) run_power(A, v0, n, e);

```

```

17
18 disp('--- Case A ---');
19 [vA, lambdaA, iterA] = power_method([1;0;0]);
20 disp('Eigenvalue:'); disp(lambdaA);
21 disp('Eigenvector:'); disp(vA);
22 disp('Iterations:'); disp(iterA);
23
24 disp('-----');
25
26 disp('--- Case B ---');
27 [vB, lambdaB, iterB] = power_method([1;1;1]);
28 disp('Eigenvalue:'); disp(lambdaB);
29 disp('Eigenvector:'); disp(vB);
30 disp('Iterations:'); disp(iterB);
31
32 % ----- FUNCTION -----
33 function [v, lambda, k] = run_power(A, v, n, e)
34     for k = 1:n
35         w = A * v;
36         lambda = max(abs(w));
37         v_new = w / lambda;
38
39         if norm(v_new - v) < e
40             v = v_new;
41             return;
42         end
43
44         v = v_new;
45     end
46 end

```

Listing 3: Task 02 — Power Method for 3×3 matrix (both cases)

► *Output*► *ComparisonTable*

Case	Initial Guess	λ^*	v_1^*	v_2^*	v_3^*	Iterations
A	[1; 0; 0]	6.3723	0.8431	1.0000	0.8431	12
B	[1; 1; 1]	6.3723	0.8431	1.0000	0.8431	12

Table 3: Comparison of Cases A and B for Task 02

Observation

Both Case A ($v^{(0)} = [1; 0; 0]$) and Case B ($v^{(0)} = [1; 1; 1]$) converge to the **same dominant eigenvalue** $\lambda^* \approx 6.3723$ with eigenvector **[0.8431; 1.0000; 0.8431]**. Both cases reach the iteration limit of 12 without triggering early termination via the tolerance condition, indicating that the 3×3 matrix requires more iterations to converge to $\varepsilon = 10^{-5}$ compared to the 2×2 case. The symmetric eigenvector

structure ($v_1 = v_3$) reflects the symmetric structure of the matrix A . The choice of initial guess does not affect the final result here, confirming that the dominant eigenvector is unique for this matrix.

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This lab successfully demonstrated the implementation and analysis of the **Power Method** for computing the dominant eigenvalue and eigenvector of square matrices in MATLAB. Key findings are summarized below:

- **Task 1** — 2×2 matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$: The Power Method converges to the dominant eigenvalue $\lambda^* \approx \mathbf{3.6184}$ with eigenvector $[\mathbf{0.6182}; \mathbf{1.0000}]$, matching the analytical result. Early termination (before the 10-iteration limit) confirms rapid convergence due to the well-separated eigenvalues.
- **Task 2** — 3×3 matrix $A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}$: Both initial guesses $[1; 0; 0]$ and $[1; 1; 1]$ converge to the **same dominant eigenvalue** $\lambda^* \approx \mathbf{6.3723}$ with eigenvector $[\mathbf{0.8431}; \mathbf{1.0000}; \mathbf{0.8431}]$. The symmetric structure of the matrix is reflected in the symmetric components of the eigenvector. Both cases ran to the full 12 iterations, indicating that higher-dimensional matrices generally require more iterations.
- **General observations:** The Power Method is simple to implement and effective for finding the dominant eigenpair. Its convergence rate depends on the ratio $|\lambda_2/\lambda_1|$; a smaller ratio leads to faster convergence. The choice of initial guess influences the convergence path but not the final result for matrices with a unique dominant eigenvalue.